PULSE · Phase 0 Report: Mission Analysis and Mission Definition Review

Oct 23, 2026 · Giuliano Pennacchio · Giuseppe Parisi

PULSE is a 16U CubeSat mission in very low Earth orbit (302 km) that, on request of the Italian Civil Protection, images an area hit by a disaster by day and by night and delivers georeferenced damage vectors, with an image chip for each alert, within 48 h. This report closes Phase 0 (ECSS-M-ST-10). It records the user need, the mission objectives, the trade-offs, two candidate mission concepts, the preliminary mission requirements, the latency budget and the main risks. The key outcome is that the 48 h response time is the driver of the whole mission: a first-order coverage estimate shows that a single satellite is at risk of not meeting it, so Phase 0 carries forward two candidates, a single-satellite demonstrator (C1) and a constellation of four satellites in one plane (C2). The selection between them is made in Phase A with an orbit propagator.

1. Purpose and approach

Phase 0 answers one question: is the mission needed, and by whom? The work started from the user and not from the payload. The reference solution is EarthNext (La Marca et al., CEAS Space Journal), a 16U CubeSat for Earth observation of Italy at 302 km with electric propulsion. For every element we asked whether PULSE needs more, less or something different than EarthNext.

2. User needs and stakeholders

The reference user is the Italian Civil Protection. The benchmark services are the Copernicus Emergency Management Service (Rapid Mapping) and the International Charter "Space and Major Disasters", which deliver image-based maps typically within one to a few days.

Use cases

Use case	Scenario	What the user needs to know
Earthquake in the Apennines	M6 event at 03:00. By the next day the Civil Protection wants to know which bridges on the main roads collapsed and which villages are cut off.	Collapsed bridges, roads blocked by debris or landslides, extended collapses; power outage areas at night

Use case	Scenario	What the user needs to know
Flood or landslide	Heavy rain on a valley for two days. Rescue teams need the flooded area and the interrupted roads.	Flooded areas, landslides on roads; usually under cloud cover
Wildfire at night	A fire front spreads in the evening near inhabited areas.	Active fire front, burnt area, villages left without power

Key numbers

Need	Value
Response time	Map delivered within 48 h of the request
Area	All of Italy, 5°E–20°E, 35°N–48°N (same as EarthNext)
Level of detail	Infrastructure and extended collapses (bridges, roads, blocks), not single buildings: a collapsed bridge is visible at 2–3 m, single-building damage needs ⩽ 0.5 m
Day and night	Day: physical damage. Night: fires and blackouts, which show where normal life has stopped
Product	Georeferenced vectors (GIS format, e.g. GeoJSON) plus an image chip for each alert, readable on a tablet in the field

Stakeholders

Stakeholder	Role
Italian Civil Protection	User: sends requests, receives and uses the products
PULSE control centre	Operator: plans acquisitions, commands the spacecraft, forwards the products
Ground station providers	Contacts for commanding and downlink (high-latitude and national stations)

Stakeholder	Role
Reference data providers	OpenStreetMap (road and bridge vectors), digital elevation model (DEM)
Launch provider	Release into the operational sun-synchronous plane
Frequency and space regulators	Frequency coordination (ITU and national authority), debris mitigation rules

3. Mission objectives

Type	Objective
Primary (technology)	Demonstrate in VLEO the chain "on-demand pointing → on-board AI processing → vector delivery" on a 16U CubeSat.
Application 1	Deliver to the Civil Protection vector maps of damage to infrastructure, floods and fires within 48 h of the request.
Application 2	Demonstrate night-time observation of fires and blackouts from a CubeSat.
Secondary	Demonstrate fast off-nadir pointing up to 45° with forward motion compensation (FMC).

Mission success criteria

Level	Criterion
Minimum success	At least one satellite operates at ~302 km and delivers, within 48 h of a request, a vector product with image chip on a test target, with both the day and the night channel demonstrated.
Full success	The mission requirements of Section 7 are met over the 2-year operational lifetime, including MR-02.

4. Trade-offs

4.1 Resolution and platform

Two first-order relations set what a 16U can see from 302 km: the ground sampling distance and the diffraction limit.

$GSD = h \cdot p / f$ $\delta diff = 1.22 \cdot \lambda \cdot h / D$

Example. For a GSD of 0.5 m with 5 µm pixels the focal length must be 302 km × 5 µm / 0.5 m ≈ 3.0 m. To resolve 0.5 m at λ = 0.55 µm the aperture must be ≈ 40 cm, while the cross-section of a 16U is about 22–23 cm. With a 12 cm aperture the diffraction limit is 1.7 m: sub-metre detail is not physically reachable on a 16U at this altitude.

Option	Gain	Cost
16U at ~1–1.5 m GSD, focus on roads and bridges **(selected)**	Stays a CubeSat, consistent with EarthNext	No single-building damage
Lower orbit (~250 km)	Better resolution	Much stronger drag, more propellant, shorter life
Microsatellite (50–150 kg) for < 0.5 m	Sub-metre possible	No longer a CubeSat; cost and complexity rise; EarthNext heritage lost

A PAN GSD below 1 m with a 12 cm aperture was also rejected: the real detail would stay at ~2 m (diffraction), while halving the GSD (for example from 1 m to 0.5 m) gives 4× less light per pixel and 4× more data.

4.2 Payload sensors

Option	Assessment
Silicon only: CMOS day + sCMOS night **(selected)**	Day PAN + RGB-NIR, night low-light panchromatic; no cooling; sees fires and blackouts
CMOS + InGaAs SWIR	Loses blackout detection; needs cooling, more power and volume
Three optical sensors	Does not fit volume and power of a 16U

Option	Assessment
Micro-SAR on 16U	Sees through clouds, but ~5–6 m resolution, large antenna, very high power, low maturity; does not see fires or blackouts

4.3 Orbit

The orbit is taken from EarthNext: mean altitude 302 km (mean semi-major axis 6680.2 km), ground-track repeat of 159 revolutions in 10 days, frozen eccentricity 0.0011 with argument of perigee 90°, sun-synchronous with LTDN 10:30. The nominal SSO inclination is 96.68°; an injection value of 96.8° is used as a bias to compensate the solar-induced inclination drift (about −0.33°/year). Over Italy the satellites pass around 10:30 (day) and 22:30 (night) local time: the mid-morning pass gives shadows that help to spot collapses, and the night pass gives the second imaging opportunity. A 51.6° non-sun-synchronous orbit was rejected: the local time of the passes drifts, so lighting conditions over Italy change over a cycle of about two months and night imaging is not guaranteed.

4.4 One satellite or a constellation

The natural starting point is a single 16U demonstrator, like EarthNext. To check it against the 48 h response time we estimated how often one point in Italy can be imaged.

First-order estimate. With 159 revolutions in 10 days, adjacent ground tracks are 252 km apart at the equator and about 187 km apart at 42°N. At 30° off-nadir the satellite reaches targets up to ~176 km from its track (minimum elevation 58.4°), i.e. a strip ~350 km wide; at 45° up to ~310 km (elevation 42.2°). Each point is therefore seen about twice per 10-day cycle on the day pass and twice on the night pass.

Satellites (same plane)	Accesses per 10 days, 30°	Mean wait, 30°	Accesses per 10 days, 45°	Mean wait, 45°
1	~4	~65 h	~7	~36 h
2	~7	~32 h	~13	~18 h
3	~11	~22 h	~20	~12 h
4	~15	~16 h	~26	~9 h

Day + night accesses. Means assume evenly spread passes; in reality the repeating ground track groups them, so the longest waits are several times the mean. Requirement MR-02 refers to 90% of the requests, so the quantity that matters is close to the worst case, not the mean: it can only be obtained with an orbit propagator.

With one satellite the mean wait at 30° (~65 h) already exceeds the whole 48 h response time, and even at 45° it uses most of it. This is a warning, not yet a proof: the estimate is coarse and the real distribution of the waits is not known.

Option	Gain	Cost
Single satellite, relaxed requirement (e.g. 96 h)	One spacecraft, simplest mission	Much lower value for first responders
Single satellite, 45° as nominal	About twice the accesses	Day resolution degrades to ~4 m, at the limit for bridges; still little margin
2–3 satellites	Mean wait roughly halved or better	Worst cases may still be close to or above 48 h
4 satellites in one plane, 90° apart	Mean wait ~16 h at 30°; one launch; identical spacecraft and orbit	Four spacecraft; the 90° spacing must be built after release
Two orbital planes	More regular revisit	Two launches (a plane change is not affordable on a 16U); one plane leaves the 10:30 local time and EarthNext heritage

> **Outcome of Phase 0.** Two candidate concepts are carried into Phase A: **C1**, a single 16U demonstrator as EarthNext, and **C2**, a constellation of four identical 16U CubeSats in the same sun-synchronous plane, spaced 90° in mean anomaly. The first-order estimate favours C2. The selection is made in Phase A (step A1) with an access and latency analysis in GMAT, against MR-02.

4.5 The "before" reference on board

Change detection compares "before" and "after". Italy covers about 300,000 km²: at 1 m that is about 300 billion pixels, hundreds of GB. It does not fit on board and cannot be uplinked (EarthNext receives 112 kbps in S-band).

Option	Assessment
Stored images of high-risk areas	Large memory, covers only part of the territory

Option	Assessment
OpenStreetMap vectors + DEM **(selected)**	Roads and bridges for all of Italy in a few MB; a 30 m DEM for Italy is ~670 MB. The AI checks "is this road still there and passable?"
Previous pass of the same satellite	Useful as a complement; depends on clouds and on recent passes

4.6 Clouds, night and downlink

An optical payload cannot see through clouds. The scope is kept to all event types (infrastructure damage, floods, fires) with the limit declared openly: the night pass gives a second chance to find a clear sky. At night a dark area can be a blackout or a cloud: a blackout alert must be confirmed, not "possible", so the method to separate the two is an open point. This is different from MR-08, where changes in artificial lights only flag possible damage to be checked by day.

The product drives the downlink. A 10 × 10 km PAN image at 1.5 m is about 44 million pixels, hundreds of MB with all bands; the vectors of 20 bridges and 50 road segments are a few kB and pass in a fraction of a second even in S-band (EarthNext transmits at 2 Mbps: 50 kB ≈ 0.2 s). PULSE downlinks vectors and light image chips, never full images.

5. Candidate mission concepts

The two candidates share the same spacecraft, payload and orbit; they differ only in the number of satellites.

Element	Common baseline (C1 and C2)
Spacecraft	16U CubeSat derived from EarthNext
Orbit	302 km mean altitude, inclination 96.8° (biased), LTDN 10:30, e = 0.0011, ω = 90°; altitude kept by electric propulsion
Propulsion	Iodine electric thruster as EarthNext (0.5 mN, Isp up to 550 s, 84 W peak)
Optics	Three-mirror anastigmat (TMA, no lenses), aperture ~12 cm, focal length ~0.7 m (TBC)
Day channel	CMOS with PAN (GSD ~1–1.5 m) and RGB-NIR (GSD ~3–5 m); pansharpening only for the image chips

Element	Common baseline (C1 and C2)
Night channel	Dual-gain sCMOS panchromatic with FMC and binning (~10–15 m on towns, ~3–5 m on fires)
On-board AI	Fast orthorectification with attitude, orbit and DEM; comparison with OSM vectors; extraction of alerts and image chips
Ground segment	High-latitude station(s) for frequent contacts (e.g. Svalbard) and the Fucino station for national commanding; final set TBC
Off-nadir	Nominal quality up to 30°, degraded pointing up to 45°
Lifetime	≥ 2 years of operations; natural re-entry from 302 km at end of life
Test targets	Aosta, Gemona, Napoli, Norcia, Palermo (north to south, mountains, coast, large city)

	C1 – Single demonstrator	C2 – Constellation of 4
Space segment	1 × 16U	4 × 16U in one plane, 90° apart in mean anomaly
Launch	One CubeSat rideshare to SSO 10:30	One launch releasing four satellites in the same plane
Revisit (first-order)	Mean wait ~65 h at 30°, ~36 h at 45°	Mean wait ~16 h at 30°, ~9 h at 45°
Strengths	Simplest, cheapest, direct EarthNext heritage	Likely to meet 48 h; redundancy if one satellite fails
Weaknesses	48 h probably not met	Four spacecraft; spacing to be built after release; higher cost and operations workload
To verify in Phase A	Real distribution of the waits (90th percentile)	Same, plus how to build and keep the 90° spacing, ΔV, launch availability

6. Emergency scenario and latency budget

Example: earthquake in the Apennines at 03:00. The Civil Protection sends the request to the PULSE control centre. At the first contact the target (coordinates and event type) is uploaded to the satellite that will pass first. That satellite rotates towards the target, activates FMC and images it with the day or night channel. On board, the image is orthorectified, compared with OSM and turned into alerts and image chips. These are downlinked at the first useful contact, and the control centre forwards the GIS file to the Civil Protection.

Step	Allocation	Driver
Request → control centre	2 h	Procedures with the Civil Protection
Command upload	12 h (TBC)	Number of ground stations (S-band/UHF)
Wait for a useful pass	24 h (TBC)	Number of satellites, off-nadir limit, day/night, clouds
Imaging + AI processing	1 h (TBC)	AI board performance
Wait for downlink contact	6 h (TBC)	Number of ground stations
Delivery to the user	1 h	Ground network
Total	**46 h (2 h margin on 48 h)**	

All values are allocations, not results. The budget closes on paper with a small margin. The 24 h allocation for the wait is the critical item: for C1 the first-order estimate (mean wait ~65 h at 30°) does not support it, while for C2 the mean wait (~16 h) fits but the 90th percentile is unknown. Both must be verified in Phase A with an access and contact analysis. If the day pass is cloudy, the night pass is the next opportunity: the night channel is essential to meet 48 h.

7. Preliminary mission requirements

Values marked TBC are confirmed by the Phase A analyses.

ID	Requirement	Verification
MR-01	PULSE shall be able to observe any point of the Italian territory (5°E–20°E, 35°N–48°N).	Analysis (GMAT)
MR-02	PULSE shall deliver vectors and image chips within 48 h of the request in 90% of the requests (TBC).	Analysis + simulation
MR-03	By day, at nadir, PULSE shall achieve an effective resolution ⩽ 2 m in panchromatic and ⩽ 5 m in multispectral (RGB-NIR).	Optical analysis, then test
MR-04	By night, PULSE shall achieve an effective resolution ⩽ 15 m on lit areas and ⩽ 5 m on fires (TBC).	Radiometric analysis (SNR)
MR-05	PULSE shall provide nominal image quality up to 30° off-nadir and shall be able to point up to 45° off-nadir in degraded mode.	Analysis + GNC simulation
MR-06	By day PULSE shall detect collapsed bridges, landslides on roads, extended collapses, flooded areas and fires (smoke and burnt areas).	AI test on datasets
MR-07	By night PULSE shall detect fires and blackouts, distinguishing a blackout from cloud cover (method TBD).	Analysis + AI test
MR-08	By night PULSE shall flag possible infrastructure damage from changes in artificial lights.	AI test
MR-09	The product for each alert shall be a georeferenced vector (GIS format, e.g. GeoJSON, TBC) with an image chip.	Inspection
MR-10	The on-board AI shall orthorectify images using attitude, orbit and DEM data and compare them with preloaded OSM vectors.	Test
MR-11	Each PULSE satellite shall operate in a sun-synchronous orbit at ~302 km with LTDN 10:30, with altitude maintained by electric propulsion.	Analysis
MR-12	Each PULSE satellite shall have an operational lifetime of at least 2 years.	Analysis
MR-13	Each PULSE satellite shall be a 16U CubeSat compatible with standard deployers.	Inspection

ID	Requirement	Verification
MR-14	At end of life each PULSE satellite shall re-enter the atmosphere in compliance with space debris mitigation rules.	Analysis

8. Constraints, critical technologies and programmatics

Constraints

Constraint	Effect on PULSE
CubeSat standard (16U)	Volume, mass (EarthNext max 35.6 kg) and power limits; standard deployer interface
Launch	Rideshare or dedicated release into the 10:30 sun-synchronous plane; availability of such launches
Frequencies	S-band (and possibly UHF) links need frequency coordination (ITU filing, national authority)
Space debris	Re-entry within the allowed time after end of life; at 302 km natural re-entry is fast
Coverage area	Italy only: products, ground network and reference data sized for 5°E–20°E, 35°N–48°N

Critical technologies (preliminary)

Technology	Why it is critical	Preliminary maturity
On-board AI: orthorectification + change detection against OSM	Core of the mission; false alarms and processing time drive the product	Low: concept and lab tests (TRL ~3–4)
Night sCMOS channel with FMC	Night SNR at 302 km and FMC on a CubeSat are not yet demonstrated together	Low to medium (TRL ~4)
TMA with ~0.7 m focal length in the 16U volume	Must fit in about 8U as the EarthNext imager	Medium (TRL ~5)

Technology	Why it is critical	Preliminary maturity
VLEO operations with iodine electric propulsion	Continuous drag compensation at 302 km for 2 years	Medium to high: iodine thrusters have flown; EarthNext design heritage (TRL ~6–7)
Fast off-nadir pointing up to 45°	Short imaging windows; reaction wheel sizing	Medium (TRL ~5–6)

The TRL values are first estimates to be checked with a market survey in Phase A.

Programmatic aspects

Item	Phase 0 assessment
Development logic	ECSS phases 0 → A (feasibility, PRR) → B (preliminary definition, SRR/PDR) → C (detailed definition, CDR)
Main cost drivers	Number of spacecraft (C1 or C2), launch, ground station services, payload and AI development
Cost estimate	Order of magnitude to be set in Phase A, once the concept is selected
Heritage	EarthNext platform, propulsion and orbit; reduces the development risk of the bus

9. Main risks

ID	Risk	Prob.	Impact	Mitigation to study in Phase A
R-01	The TMA with ~0.7 m focal length does not fit in the 16U volume	Medium	High	Folded optics; comparison with the 8U allocated to the EarthNext imager
R-02	Night signal-to-noise ratio too low	Medium	High	Near-full FMC, binning, dual-gain sCMOS; radiometric budget

ID	Risk	Prob.	Impact	Mitigation to study in Phase A
R-03	A blackout cannot be distinguished from a cloud	High	High	Method to be found (previous nights, meteo data, moonlight, revisit)
R-04	Too many AI false alarms (shadows, seasons, construction sites)	Medium	High	Tests on public disaster datasets; confidence thresholds
R-05	Inaccurate off-nadir orthorectification in mountains	Medium	Medium	DEM on board; accurate attitude knowledge
R-06	45° slews and night FMC exceed the reaction wheels	Medium	Medium	GNC simulation; 45° is already a degraded mode
R-07	Not enough power for AI + thruster + manoeuvres	Medium	Medium	Separate AI and thrusting in time; power budget
R-08	48 h not met because of clouds or too few contacts	Medium	Medium	Night pass; high-latitude stations; more satellites (C2)
R-09	The AI board overheats	Low	Medium	Thermal analysis; pulsed processing
R-10	A single satellite cannot meet MR-02 (first-order mean wait ~65 h at 30°)	High	High	Access and latency analysis in GMAT (A1); constellation C2; relaxed requirement as last resort

10. Open points for Phase A

- Selection between C1 and C2: access and latency analysis with an orbit propagator, mean and 90th percentile wait at 30° and 45°, day only and day + night, including the clustering of passes.
- Selection of the ground stations and confirmation of the latency budget.
- Final wording of MR-02 (percentage, off-nadir allowed by day).
- If C2: how to release the four satellites and build the 90° spacing, and launch availability.
- ΔV and propellant for 2 years, including orbit lowering after release (EarthNext: > 275 m/s for 3 years).
- Exact focal length, pixel size and GSD of the two channels.

- Night radiometric budget (SNR) on cities, villages and fires.
- Method to distinguish blackouts from clouds at night (previous nights, meteo data, moonlight).
- Memory needed for OSM vectors and DEM.
- Market survey for the critical technologies and first cost estimate.

11. MDR checklist and conclusion

Review question	Status	Where
Is the user need clear and written with a concrete example?	Yes	Section 2
Are the objectives and the success criteria defined?	Yes	Section 3
Does every choice have its rationale, including the rejected alternatives?	Yes	Section 4
Are there 2–3 candidate concepts to compare in Phase A?	Yes: C1 and C2	Section 5
Does the latency budget fit within 48 h?	On paper (2 h margin); the wait allocation must be verified in Phase A	Section 6
Does every requirement have an ID and a verification method?	Yes	Section 7
Are constraints and critical technologies identified?	Yes, preliminary	Section 8
Do the high risks have a path to be reduced?	Yes	Sections 9–10

> **Phase 0 closed.** The mission need is confirmed: the Civil Protection needs damage maps of Italy within 48 h, by day and by night, at the level of roads and bridges. Two candidate concepts share the same 16U spacecraft, day and night optical payload, on-board AI and EarthNext orbit: a single demonstrator (C1) and a constellation of four satellites in one plane (C2). The 48 h response time is the main driver, and the first-order estimate favours C2. The project can proceed to Phase A, where the access and latency analysis selects the concept, together with the open points of Section 10.